RESEARCH ARTICLE

Rate of Autoimmune Diseases in the 10 Years Preceding Myasthenia Gravis Diagnosis

A Nationwide Cohort Study in South Korea

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Abstract

Background and Objectives

Patients with myasthenia gravis (MG) are frequently associated with other autoimmune diseases. However, the temporal relationship between autoimmune diseases and MG remains unclear. This study aimed to evaluate the rates and temporal patterns of autoimmune diseases during the 10 years preceding MG diagnosis.

Methods

This retrospective population-based cohort study analyzed claims data from the Korean National Health Insurance Service between 2010 and 2021 and included individuals aged ≥20 years. MG was defined using the International Classification of Diseases, 10th Revision (ICD-10) code (G70.0) and Rare Intractable Disease registration (V012). Each MG case was matched with 10 controls by age, sex, and index year. Outcomes were autoimmune thyroid disease (AITD), systemic lupus erythematosus (SLE), seropositive rheumatoid arthritis (SRA), Sjögren syndrome (SjS), psoriasis, type 1 diabetes mellitus (T1DM), Crohn disease, and ulcerative colitis occurring during the 10 years before the index year. Rate ratios (RRs) and 95% CIs were estimated using Poisson regression models for 0–2, 2–5, and 5–10 years before the index year.

Results

A total of 8,355 patients with MG and 83,550 matched controls were included. The mean age (SD) was 53.7 (16.0) years, and 44.1% were male. Patients with MG had a higher rate of any autoimmune disease during 10 years before diagnosis than controls (RR 2.12, 95% CI 1.97–2.28), with the highest rates in 0–2 years before diagnosis (RR 4.27, 95% CI 3.78–4.83). SLE (RR 4.87, 95% CI 2.71–8.77), SjS (RR 4.75, 95% CI 3.01–7.50), AITD (RR 3.66, 95% CI 3.29–4.07), and SRA (RR 2.45, 95% CI 1.89–3.19) showed the strongest associations. Psoriasis (RR 1.68, 95% CI 1.25–2.25) and T1DM (RR 2.03, 95% CI 1.52–2.71) were associated with MG only within 0–2 years before diagnosis, whereas Crohn disease and ulcerative colitis were not associated.

Discussion

Autoimmune diseases were more frequent in the years preceding MG diagnosis than in controls, particularly within 2 years before diagnosis. Strong associations were observed for SLE, SjS, AITD, and SRA. However, these findings should be interpreted with caution because of limitations of administrative claims data.

Introduction

Myasthenia gravis (MG) is a chronic autoimmune neuromuscular disorder characterized by fluctuating skeletal muscle weakness, mainly caused by autoantibodies targeting components of the neuromuscular junction, most commonly the acetylcholine receptor (AChR).[1] MG is

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Glossary

AChR = acetylcholine receptor; **AITD** = autoimmune thyroid disease; **ICD-10** = International Classification of Diseases, 10th Revision; **MG** = myasthenia gravis; **NHIS** = National Health Insurance Service; **NHSP** = National Health Screening Program; **OR** = odds ratio; **RA** = rheumatoid arthritis; **RID** = Rare Intractable Disease; **RR** = rate ratio; **SjS** = Sjögren syndrome; **SLE** = systemic lupus erythematosus; **SRA** = seropositive RA; **T1DM** = type 1 diabetes mellitus.

driven by autoreactive T-cell–dependent B-cell activation, leading to pathogenic autoantibody production, with thymic abnormalities and genetic susceptibility contributing to disease development.[2-5]

Patients with MG are more likely to develop other autoimmune diseases compared with the general population, with reported prevalences ranging from 13% to 22% across different cohorts.[6,7] This increased risk varies according to demographic and clinical subtypes of MG. Women and individuals with early-onset MG (onset age <50 years) show a markedly higher frequency of autoimmune comorbidities.[7-9] In contrast, patients with late-onset MG or thymoma-associated MG tend to exhibit lower rates of additional autoimmune disorders, suggesting a distinct immunopathologic background.[10]

Among the autoimmune diseases associated with MG, autoimmune thyroid diseases (AITD) are consistently the most common, followed by systemic lupus erythematosus (SLE), rheumatoid arthritis (RA), and type 1 diabetes mellitus (T1DM).[11] However, data on autoimmune diseases preceding MG onset remain limited. A nationwide population-based study from Taiwan reported that individuals with Hashimoto thyroiditis or Graves' disease had an increased risk of developing MG, with adjusted odds ratios (ORs) of 2.87 (95% CI 1.18–6.97) and 3.97 (95% CI 2.71–5.83), respectively.[12] Although associations between MG and SLE or RA have been described,[13,14] epidemiologic studies are scarce. Because coexisting autoimmune diseases may influence clinical features, prognosis, and therapeutic decisions, it is clinically meaningful to clarify these associations.

In this study, we investigated the association between MG and preceding autoimmune diseases and evaluated temporal risk patterns of autoimmune disease occurrence during the 10 years before MG diagnosis compared with age- and sex-matched controls.

Methods

Data Source

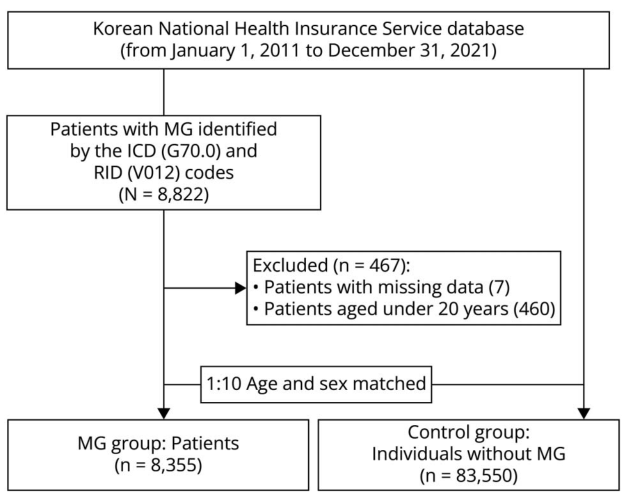
We conducted a retrospective, population-based matched cohort study using data from the Korean National Health Insurance Service (NHIS). The Korean NHIS is a mandatory insurance program that has provided universal coverage since 1989. It covers approximately 97% of the Korean population, whereas the remaining 3% is covered by the Medical Aid program, which is also linked to the NHIS database. The NHIS database includes demographic information, diagnostic codes, medical procedures, prescription records, and health care utilization data. Diagnoses are coded according to the International Classification of Diseases, 10th Revision (ICD-10), and the NHIS is linked to the national mortality database.[15]

In addition, the NHIS incorporates data from the National Health Screening Program (NHSP), which provides biennial examinations to all adults aged 40 years and older and to employees aged 20 years and older.[16] The program includes anthropometric and laboratory measurements as well as self-reported health-related behaviors, enabling adjustment for baseline health status and lifestyle factors.[17]

Study Population

We constructed a cohort of patients diagnosed with MG using data from the NHIS database between January 1, 2011, and December 31, 2021 (Figure 1). MG was defined by the concurrent presence of both the ICD-10 code for MG (G70.0) and registration in the Rare Intractable Disease (RID) program (V012). The RID program is a national registry that requires diagnostic confirmation by a board-certified specialist, ensuring accurate identification of patients with MG.[18] Because RID registration substantially reduces out-of-pocket medical expenses, nearly all eligible patients with MG are enrolled.

Figure 1 Flowchart of Study Population Enrollment



ICD = International Classification of Diseases; MG = myasthenia gravis; RID = rare intractable disease.

The index year was defined as the calendar year in which both the ICD-10 and RID codes were first recorded. Patients younger than 20 years who had not undergone NHSP health screening and those with incomplete demographic information were excluded.

For each patient with MG, 10 control individuals without MG were randomly selected from the general population and matched 1:10 by age, sex, and index year, with controls assigned the same index year as their corresponding cases.

Definition of Autoimmune Diseases

The primary outcomes were the diagnosis of autoimmune diseases before the index year, including SLE, seropositive RA (SRA), Sjögren syndrome (SjS), AITD, psoriasis, T1DM, Crohn disease, and ulcerative colitis. These conditions were selected based on their relatively high prevalence in Korea, immunopathogenic relevance to MG, and the feasibility of reliable case identification using claims data.[10,19]

Autoimmune diseases were identified using relevant ICD-10 codes (Table 1). For diseases covered by the RID program, diagnoses were defined by the concurrent presence of the corresponding ICD-10 and RID codes. For diseases without an RID code, diagnoses required at least 2 outpatient visits (with events captured at the second visit) or 1 hospitalization with the relevant ICD-10 code, to improve diagnostic validity and minimize misclassification.

Covariates

Baseline covariates were assessed in the index year. Demographic factors included age, sex, residential area, and income level. Residential areas were classified as urban or rural based on administrative district codes. Income status was defined as the lowest quartile of insurance premium contributors or registration as a medical aid beneficiary.

Comorbidities were defined using administrative claims and health screening records at index year. Type 2 diabetes mellitus (T2DM) was defined as at least 1 annual claim with ICD-10 codes E11–E14 and a prescription for antidiabetic medication or a fasting blood glucose level ≥126 mg/dL. Hypertension was defined as at least 1 annual claim with ICD-10 codes I10–I13 or I15 and a prescription for antihypertensive medication or systolic/diastolic blood pressure ≥140/90 mm Hg. Dyslipidemia was defined as at least 1 annual claim with ICD-10 code E78 and a prescription for lipid-lowering therapy or total cholesterol ≥240 mg/dL. Thymic disease was defined as a diagnosis of malignant neoplasm of the thymus (C37), benign neoplasm of the thymus (D15.0), or thymic hyperplasia (E32.0).

Statistical Analysis

Using this matched cohort, we assessed whether patients with MG had an increased likelihood of autoimmune disease diagnoses during the 10 years preceding the index year. The prediagnostic period was divided into 3 intervals (0–2, 2–5, 5–10 years), with additional annual analyses performed to examine year-by-year trends.

Baseline characteristics were summarized using means with standard deviations for continuous variables and counts with percentages for categorical variables and compared using the Student *t* test or χ^2 test, as appropriate. Poisson regression models were used to estimate rate ratios (RRs) and 95% CIs for each autoimmune disease, comparing patients with MG to matched controls. Models were conditioned on the matching variables (age, sex, and index year) and adjusted for income, residential area, and comorbidities (hypertension, T2DM, dyslipidemia, and thymic disease). RRs were estimated for the overall 10-year period, each prediagnostic interval, and individual years when applicable. To further validate the results, 1:5 propensity score matching was additionally performed for the prediagnostic interval analyses. Subgroup analyses were conducted according to age at index year (<50 vs ≥50 years)[20] and the presence of thymic disease.

Table 1 Operational Definitions of Autoimmune Diseases

Autoimmune diseases	ICD-10 code	RID code	Additional conditions
Systemic lupus erythematosus	M32.1	V136	N/A
Seropositive rheumatoid arthritis	M05	V223	N/A
Sjögren syndrome	M35.0	V139	N/A
Autoimmune thyroid diseases	E05.0, E06.3	N/A	At least 2 outpatient visits or 1 hospitalization
Psoriasis	L40	N/A	At least 2 outpatient visits or 1 hospitalization
Type 1 diabetes mellitus	E10.9	N/A	At least 2 outpatient visits or 1 hospitalization
Crohn disease	K50	V130	N/A
Ulcerative colitis	K51.9	V131	N/A

Abbreviations: ICD-10 = International Classification of Diseases, 10th Revision; N/A = not applicable; RID = rare intractable diseases.

All analyses were performed using SAS version 9.4 (SAS Institute Inc., Cary, NC), with 2-sided p values <0.05 considered statistically significant.

Standard Protocol Approvals, Registrations, and Patient Consents

The study protocol was approved by the Institutional Review Board of Samsung Medical Center (IRB No. SMC 2024-05-037), and the requirement for informed consent was waived because only deidentified data were analyzed. The analyses were performed using data from the NHIS-Big Data Platform (NHIS-2025-10-1-070), for which review and access approval were granted by the Korean NHIS.

Data Availability

The data that support the findings of this study are available from the Korean NHIS but are not publicly available due to restrictions. Data are available from the corresponding author upon reasonable request and with permission of the NHIS.

Results

Baseline Characteristics of the Study Population

A total of 8,355 patients with MG and 83,550 matched controls were included (Figure 1). The mean age at the index year was 53.7 ± 16.0 years, and 44.1% were male in both groups. The time interval between first ICD code and RID registration for MG was median (interquartile range) 14 (0–51) days (eFigure 1).

Compared with controls, patients with MG had higher prevalences of T2DM (14.7% vs 10.6%), hypertension (35.3% vs 27.1%), dyslipidemia (30.5% vs 21.0%), and thymic disorders (10.07% vs 0.01%; all $p < 0.001$). Urban residence was more common among patients with MG than controls (47.6% vs 45.1%, $p < 0.001$), whereas low-income status did not differ significantly (Table 2).

RR of Autoimmune Diseases Before the Index Year

During the 10-year prediagnostic period, patients with MG showed a significantly increased rate of any autoimmune disease compared with matched controls (RR 2.12, 95% CI 1.97–2.28; Figure 2). The highest rate was observed within 0–2 years before the index year (RR 4.27, 95% CI 3.78–4.83), with elevated rates persisting during the 2–5 years (RR 1.70, 95% CI 1.47–1.96) and 5–10 years (RR 1.44, 95% CI 1.28–1.63) of intervals (Table 3).

Among individual autoimmune diseases, SLE had the highest RR over the entire 10-year period (RR 4.87, 95% CI 2.71–8.77), followed by SjS (RR 4.75, 95% CI 3.01–7.50), AITD (RR 3.66, 95% CI 3.29–4.07), and SRA (RR 2.45, 95% CI 1.89–3.19). These diseases persistently showed elevated RRs across all prediagnostic intervals. Psoriasis (RR 1.33, 95% CI 1.14–1.56) and T1DM (RR 1.21, 95% CI 1.04–1.42) each showed modest associations that were mainly confined to 0–2 years interval before the index year. Crohn disease and ulcerative colitis were not significantly associated.

Annual analyses showed elevated RRs across most of the prediagnostic period, with the highest rate observed within 1 year before the index year (RR 6.54, 95% CI 5.61–7.62; eTable 1). In particular, AITD and SLE demonstrated marked increases within 1 year before diagnosis, with RRs of 15.20 (95% CI 12.31–18.78) and 12.39 (95% CI 2.74–56.11), respectively.

In the additional 1:5 propensity score matching analysis performed to validate the results, thymic disorders were excluded because of substantial imbalance between the MG and control groups (absolute standardized difference 0.47; eTable 2). During the entire 10-year period, RR of any autoimmune disease in patients was 2.10 (95% CI 1.94–2.27; eFigure 2), consistent with the primary analysis. Similar patterns were observed across individual autoimmune diseases across all time intervals (eTable 3).

Subgroup Analysis

Across age subgroups based on age at the index year (<50 vs ≥50 years), patients with MG exhibited higher RRs of autoimmune diseases than matched controls (eFigure 3, A–I). Among individuals aged <50 years, SjS (RR 13.67, 95% CI 5.77–32.37), AITD (RR 5.89, 95% CI 5.02–6.91), and SLE (RR 5.19, 95% CI 2.47–10.94) showed the highest RRs. In those aged ≥50 years, SLE showed the highest RR (RR 4.17, 95% CI 1.57–11.03), followed by SjS (RR 3.05, 95% CI 1.70–5.46). Overall, the rate of any autoimmune disease was increased in both age groups, with RRs of 3.49 (95% CI 3.08–3.95) for <50 years and 1.68 (95% CI 1.53–1.84) for ≥50 years.

Subgroup analysis for thymic disorders was not feasible because of insufficient numbers in each subgroup for statistical analysis.

Discussion

In this nationwide population-based study, patients with MG showed significantly increased rates of autoimmune diseases during the 10 years before diagnosis, peaking in the 0–2 years. SLE, SjS, SRA, and AITD showed persistently increased rates across all prediagnostic intervals, whereas T1DM and psoriasis only showed elevations within the final 2 years. Crohn disease and ulcerative colitis were not significantly associated with MG during the prediagnostic period.

SLE and AITD showed consistently elevated RRs across all intervals, with a marked increase during the 0–2 years preceding MG diagnosis. For SLE, although absolute frequencies were low in both MG and non-MG cohorts (0.0%–0.1%), the

Table 2 Baseline Demographics of Patients With Myasthenia Gravis and the Matched Control Group at Index Year

	Control (n = 83,550)	Case (n = 8,355)	*p* Value
Age at index year			
Mean years (±SD)	53.7 ± 16.0	53.7 ± 16.0	1.000
<50 y	32,400 (38.8)	3,240 (38.8)	1.000
≥50 y	51,150 (61.2)	5,115 (61.2)	
Sex			
Male	36,810 (44.1)	3,681 (44.1)	1.000
Index year			1.000
2011	6,310 (7.6)	631 (7.6)	
2012	6,240 (7.5)	624 (7.5)	
2013	6,880 (8.2)	688 (8.2)	
2014	6,650 (8.0)	665 (8.0)	
2015	7,320 (8.8)	732 (8.8)	
2016	7,870 (9.4)	787 (9.4)	
2017	7,850 (9.4)	785 (9.4)	
2018	8,930 (10.7)	893 (10.7)	
2019	7,930 (9.5)	793 (9.5)	
2020	8,890 (10.6)	889 (10.6)	
2021	8,680 (10.4)	868 (10.4)	
Socioeconomic factors			
Low income	20,599 (24.7)	2,075 (24.8)	0.715
Urban residence	37,653 (45.1)	3,979 (47.6)	<0.001
Comorbidities			
Type 2 diabetes mellitus	8,837 (10.6)	1,231 (14.7)	<0.001
Hypertension	22,621 (27.1)	2,945 (35.3)	<0.001
Dyslipidemia	17,549 (21.0)	2,550 (30.5)	<0.001
Thymic disorders	5 (0.01)	841 (10.07)	<0.001

Values are presented as mean ± SD or number (percentage) as appropriate.

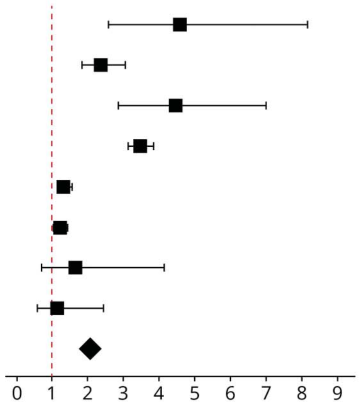
RR for SLE diagnosis was 4.87-fold higher in patients with MG. A previous single-center study from France found that 17 (1.3%) of 1,300 patients with SLE developed concomitant MG, which is higher than the 0.02% observed in the general population.[13] Among these patients, 8 developed MG after SLE diagnosis, with a median interval of 4 years (range, 0.3–13 years). These findings suggest that MG may occur several years after the diagnosis of SLE.

In AITD, the 10-year prevalence before MG diagnosis was 5.7% in patients with MG and 1.5% in those without MG, corresponding to a 3.66-fold higher RR. AITD is the most common autoimmune comorbidity among patients with MG, reported in approximately 10.1%–18.4% of patients, suggesting a strong association between the 2 disorders.[21,22] A population-based study from Taiwan comparing 1,689 patients with MG and 6,756 controls showed that Graves' disease was associated with an increased risk of MG (OR 3.97, 95% CI 2.71–5.83), as was Hashimoto disease (OR 2.87, 95% CI 1.18–6.97).[12] These findings are consistent with our results, although we did not classify AITD into Graves' disease and Hashimoto disease.

SRA was associated with elevated RRs across all intervals before MG diagnosis. However, unlike SLE or AITD, no pronounced increase in RR was observed within the 0–2-year interval. The prevalence of RA among patients with MG has been estimated at approximately 3%,[11] but epidemiologic

Figure 2 Rate Ratios* for Autoimmune Diseases During the 10-Year Period Preceding Myasthenia Gravis Diagnosis

	Control (n = 83,550)	Case (n = 8,355)	Rate ratio (95% CI)
Systemic lupus erythematosus	31 (0.04%)	18 (0.22%)	4.87 (2.71–8.77)
Seropositive rheumatoid arthritis	281 (0.34%)	70 (0.84%)	2.45 (1.89–3.19)
Sjögren syndrome	57 (0.07%)	28 (0.34%)	4.75 (3.01–7.50)
Autoimmune thyroid diseases	1,285 (1.54%)	473 (5.66%)	3.66 (3.29–4.07)
Psoriasis	1,322 (1.58)%	183 (2.19%)	1.33 (1.14–1.56)
Type 1 diabetes mellitus	1,231 (1.47%)	183 (2.19%)	1.21 (1.04–1.42)
Crohn disease	30 (0.04%)	5 (0.06%)	1.69 (0.65–4.38)
Ulcerative colitis	60 (0.07%)	7 (0.08%)	1.15 (0.52–2.52)
Any autoimmune disease	4,021 (4.81%)	892 (10.68%)	2.12 (1.97–2.28)



*Rate ratios were conditioned on the matching variables (age, sex, and index year) and adjusted for income, residential area, and comorbidities (hypertension, type 2 diabetes mellitus, dyslipidemia, and thymic disease).

studies on MG incidence among patients with RA remain limited. More recently, Mendelian randomization and transcriptomic analyses have supported evidence that RA is associated with an increased risk of developing MG (OR 1.353, 95% CI 1.081–1.693).[23]

Psoriasis and T1DM demonstrated significantly increased RRs within the 0–2 years preceding MG diagnosis, with the highest RRs observed in the year immediately before diagnosis. However, large-scale epidemiologic studies examining their associations with MG are lacking. To date, 6 cases of T1DM coexisting with MG, including pediatric and adolescent patients, have been reported.[24,25] In most cases, T1DM preceded MG, although T1DM developing after MG has also been reported. More recently, a large-scale GWAS meta-analysis integrating European and North American cohorts showed a strong genetic correlation between MG and T1DM, implicating both HLA loci (*HLA-DRB1*, *HLA-B*) and non-HLA genes (*PTPN22*, *CTLA4*) as shared susceptibility factors.[25]

Crohn disease and ulcerative colitis did not show increased RRs during the pre-diagnostic period of MG, and no specific annual intervals showed a significant increase. In contrast, a Swedish population-based study previously reported an increased risk of Crohn disease (OR 2.3, 95% CI 1.3–4.0) and ulcerative colitis (OR 2.1, 95% CI 1.3–3.5) among patients with MG.[7] Similarly, a single-center cohort study from Brazil showed an elevated prevalence of MG among patients with inflammatory bowel disease (OR 8.56, 95% CI 3.1–23.5),[26] suggesting a possible association between the 2 conditions. However, these results should be interpreted cautiously because the number of cases was small. Accordingly, larger multicenter studies are warranted to clarify whether a causal relationship exists between inflammatory bowel disease and MG.

The overlap between MG and other autoimmune diseases observed in this study is likely explained by shared pathogenic mechanisms. First, shared genetic susceptibility may play a pivotal role. Previous studies have consistently implicated HLA-DR3 and the ancestral haplotype 8.1 as common risk factors across MG, SLE, SjS, AITD, and T1DM, supporting overlapping immunogenetic backgrounds among these conditions.[27,28] In addition, the *PTPN22* polymorphism also represents a shared genetic risk factor across multiple autoantibody-driven autoimmune diseases, including MG, SLE, RA, T1DM, AITD, and psoriasis.[29,30] In contrast, it shows no significant association with ulcerative colitis and is even protective against Crohn disease. Second, dysregulated cytokine signaling and T-cell–mediated immune activation may contribute to the coexistence of MG and other autoimmune diseases.[31] The activation of proinflammatory pathways, involving Th17-related cytokines and interferon-mediated responses, has been reported across multiple autoimmune conditions, including MG, SLE, and SjS.[32-34] These pathways promote sustained immune activation and autoantibody production, which may precede overt neuromuscular symptoms and facilitate the development of MG in individuals with preexisting autoimmune activity. Third, dysregulation of immune-related microRNAs may represent a shared post-transcriptional regulatory mechanism across MG and other autoimmune diseases. Overexpression of miR-155 has been consistently reported in MG, SLE, SjS, RA, T1DM, and psoriasis, implicating impaired immune tolerance and proinflammatory T-cell activation as common features.[35,36] Likewise, increased miR-21 expression observed across MG, SLE, and RA supports shared regulatory pathways associated with sustained immune activation and autoantibody production.[37-39] Finally, environmental factors such as vitamin D deficiency, gut microbiota dysbiosis, and EBV infection may modulate immune tolerance and promote autoimmune clustering in susceptible individuals.[31]

Table 3 Occurrence and RRs[a] of Autoimmune Diseases During the 3 Intervals (0–2, 2–5, and 5–10 Years) Before the Index Year

	0–2 y		2–5 y		5–10 y	
	Control (n = 83,550)	Case (n = 8,355)	Control (n = 83,550)	Case (n = 8,355)	Control (n = 83,550)	Case (n = 8,355)
Systemic lupus erythematosus						
N (%)	5 (0.0)	5 (0.1)	9 (0.0)	4 (0.0)	17 (0.0)	9 (0.1)
RR (95% CI)	8.91 (2.55–31.20)		3.31 (1.01–10.88)		4.57 (2.02–10.34)	
Seropositive rheumatoid arthritis						
N (%)	66 (0.1)	16 (0.2)	106 (0.1)	32 (0.4)	109 (0.1)	22 (0.3)
RR (95% CI)	2.44 (1.41–4.23)		3.04 (2.04–4.53)		1.91 (1.20–3.02)	
Sjögren syndrome						
N (%)	12 (0.0)	11 (0.1)	16 (0.0)	7 (0.1)	29 (0.0)	10 (0.1)
RR (95% CI)	8.18 (3.58–18.69)		4.19 (1.70–10.19)		3.47 (1.68–7.15)	
Autoimmune thyroid diseases						
N (%)	280 (0.3)	261 (3.1)	403 (0.5)	101 (1.2)	602 (0.7)	111 (1.3)
RR (95% CI)	9.23 (7.79–10.94)		2.42 (1.95–3.02)		1.85 (1.51–2.27)	
Psoriasis						
N (%)	311 (0.4)	53 (0.6)	421 (0.5)	55 (0.7)	590 (0.7)	75 (0.9)
RR (95% CI)	1.68 (1.25–2.25)		1.24 (0.93–1.64)		1.22 (0.96–1.55)	
Type 1 diabetes mellitus						
N (%)	234 (0.3)	57 (0.7)	377 (0.5)	48 (0.6)	620 (0.7)	78 (0.9)
RR (95% CI)	2.03 (1.52–2.71)		1.04 (0.77–1.40)		1.02 (0.80–1.29)	
Crohn disease						
N (%)	4 (0.0)	1 (0.0)	11 (0.0)	0 (0.0)	15 (0.0)	4 (0.0)
RR (95% CI)	2.69 (0.30–24.48)		N/A		2.47 (0.81–7.54)	
Ulcerative colitis						
N (%)	13 (0.0)	2 (0.0)	11 (0.0)	1 (0.0)	36 (0.0)	4 (0.0)
RR (95% CI)	1.57 (0.35–7.02)		0.89 (0.11–6.94)		1.07 (0.38–3.02)	
Any autoimmune diseases						
N (%)	848 (1.0)	374 (4.5)	1,244 (1.5)	224 (2.7)	1,929 (2.3)	294 (3.5)
RR (95% CI)	4.27 (3.78–4.83)		1.70 (1.47–1.96)		1.44 (1.28–1.63)	

Abbreviations: N/A = not applicable; RR = rate ratio.
[a] RRs were conditioned on the matching variables (age, sex, and index year) and adjusted for income, residential area, and comorbidities (hypertension, type 2 diabetes mellitus, dyslipidemia, and thymic disease).

Although treatment effects were beyond the scope of the current study, patients with autoimmune diseases are likely to receive immunomodulatory therapies more frequently than controls. Nevertheless, the association with MG remained significant. This may reflect incomplete treatment responses or refractory disease in a subset of patients. In addition, proinflammatory immune activity may persist despite immunomodulatory therapy.[40-42] Therefore, even under treatment, patients with autoimmune diseases may remain in a proinflammatory state that predisposes them to MG.

The age-stratified subgroup analysis showed that patients with MG had higher relative risks of autoimmune diseases across all age groups, with stronger associations observed in younger individuals. The attenuated associations in older groups likely reflect that most autoimmune diseases are more prevalent in middle-aged or younger individuals.[43] This pattern may, in part, be explained by differences in immunopathologic mechanisms, such as reduced thymic involvement or immunosenescence.

These findings also are consistent with known differences between early-onset and late-onset MG. Early-onset MG is more strongly associated with female predominance, thymic hyperplasia, and coexisting autoimmune diseases, including autoimmune thyroid disease.[8] In addition, early-onset MG has been linked to the HLA-A1-B8-DR3 haplotype, which confers broader susceptibility to autoimmune diseases, whereas late-onset MG is associated with different HLA risk alleles.[8,44] Non-HLA genetic factors have also been implicated, with variants in immune regulatory pathways reported in both subtypes.

The temporal association between MG and preceding autoimmune diseases suggests that MG often emerges in the context of systemic immune dysregulation. In particular, the marked increase in SLE and AITD diagnoses in the year preceding MG diagnosis may reflect increased immune activation. In clinical practice, promoting awareness of early symptoms or signs of MG among rheumatologists and other clinicians may help facilitate timely referral to neurologists. Furthermore, when treating MG in the presence of coexisting autoimmune conditions, immunosuppressive regimens should be individualized to balance therapeutic efficacy for MG with safety considerations related to comorbid autoimmune diseases.

This study has several notable strengths. First, we used a nationwide, population-based cohort derived from the Korean NHIS database, encompassing all newly diagnosed cases of MG in Korea from 2011 to 2021. By applying a 1:10 age- and sex-matched control design, we included more than 90,000 individuals, ensuring substantial statistical power and population-level generalizability. Second, to enhance diagnostic specificity for MG, we defined cases using both ICD-10 and RID codes. RID registration required specialties confirmation based on standardized diagnostic criteria and is linked to reimbursement, thereby improving the accuracy of case identification. Previous studies have demonstrated the validity of ICD/RID-based definitions in cancer and inflammatory bowel disease, with high sensitivity and specificity.[45,46] Furthermore, the crude annual incidence of MG observed in our study (eTable 4) was comparable to that reported in previous Korean epidemiologic studies and showed a similar increasing trend, supporting the reliability of our case definition.[47,48] Third, the 10-year longitudinal observation period enabled detailed temporal analyses of autoimmune disease occurrence, capturing both early and late prediagnostic phases.

Several limitations should be acknowledged. First, the use of ICD-10 and RID-based operational definitions without formal validation may have led to potential misclassification. Although combining ICD and RID codes likely improves diagnostic accuracy, validation studies for MG and other autoimmune diseases remain limited, and misclassification cannot be completely excluded. Second, the index year was defined by the timing of diagnosis rather than symptom onset. Therefore, the observed increase in autoimmune disease diagnoses within the year immediately before MG diagnosis may partially reflect intensified diagnostic evaluation leading to MG detection. In addition, a time lag between ICD coding and RID registration may occur in some patients, potentially leading to temporal misclassification. Moreover, for outpatient-based definitions requiring 2 visits, events were captured at the second visit without a time restriction, potentially introducing misclassification. Third, the use of administrative claims data precluded detailed assessment of key clinical characteristics, including MG subtype (ocular vs generalized), disease severity, antibody status (e.g., AChR, MuSK), and thymic histology. These unmeasured factors may influence the patterns of autoimmune clustering and limit further stratified analyses. Fourth, surveillance bias is a potential concern because patients with autoimmune diseases tend to have more frequent clinical encounters, which may increase the likelihood of MG detection. Conversely, undiagnosed autoimmune conditions in the general population could lead to overestimation of RRs. Fifth, several autoimmune diseases, including SLE, Crohn disease, and ulcerative colitis, had low event counts, resulting in wide CIs and limited statistical precision. Finally, the study population consisted exclusively of Korean individuals, representing a relatively homogeneous East Asian cohort. Ethnic differences in MG pathogenesis, including HLA allele distributions and the prevalence of MuSK-positive MG, have been documented between Asian and Western populations.[49,50] Although our findings are likely generalizable to populations with predominantly AChR-positive MG, studies in multiethnic cohorts are needed to further evaluate the external validity of these results.

In this nationwide matched cohort, higher rates of autoimmune diseases were observed for up to 10 years before MG diagnosis, with the highest rates in the 0–2 years preceding diagnosis and the strongest associations for SLE, SjS, AITD, and SRA. These findings suggest that MG may develop in the context of prolonged systemic immune activity. Further research should assess whether specific autoimmune comorbidity patterns can serve as early indicators of MG and validate these associations in multiethnic and multicenter cohorts.

Author Contributions

S. Kwon: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. B.-s. Kim: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. K. Han: major role in the acquisition of data; study concept or design; analysis or interpretation of data. N.J. Heo: drafting/

revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. D.W. Shin: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. J.-H. Min: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data; additional contributions: supervision.

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